

3 There are four types of immunity.

- (a) Artificial immunity develops when a person is immunised by an injection.

The table gives some statements about artificial immunity.

For each statement, put **one** cross ☒ in the appropriate box, in each row, to show which statements are correct for the types of artificial immunity.

(3)

Statement	Type of artificial immunity			
	both active and passive	active only	passive only	neither active nor passive
Antibodies are injected into the person	☒	☐	☐	☐
B cells differentiate into plasma cells	☐	☐	☐	☐
Memory cells are formed	☐	☐	☐	☐

- (b) Natural active immunity can develop when a person is infected with a virus.

- (i) Describe the role of macrophages in the development of natural active immunity to a virus, following infection.

(3)

This image shows a full page of white paper with ten horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and extend across the entire width of the page. There is no text or other markings on the paper.

(ii) Explain why both T helper cells and T killer cells are needed in the immune response to a virus.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question 3 = 10 marks)
